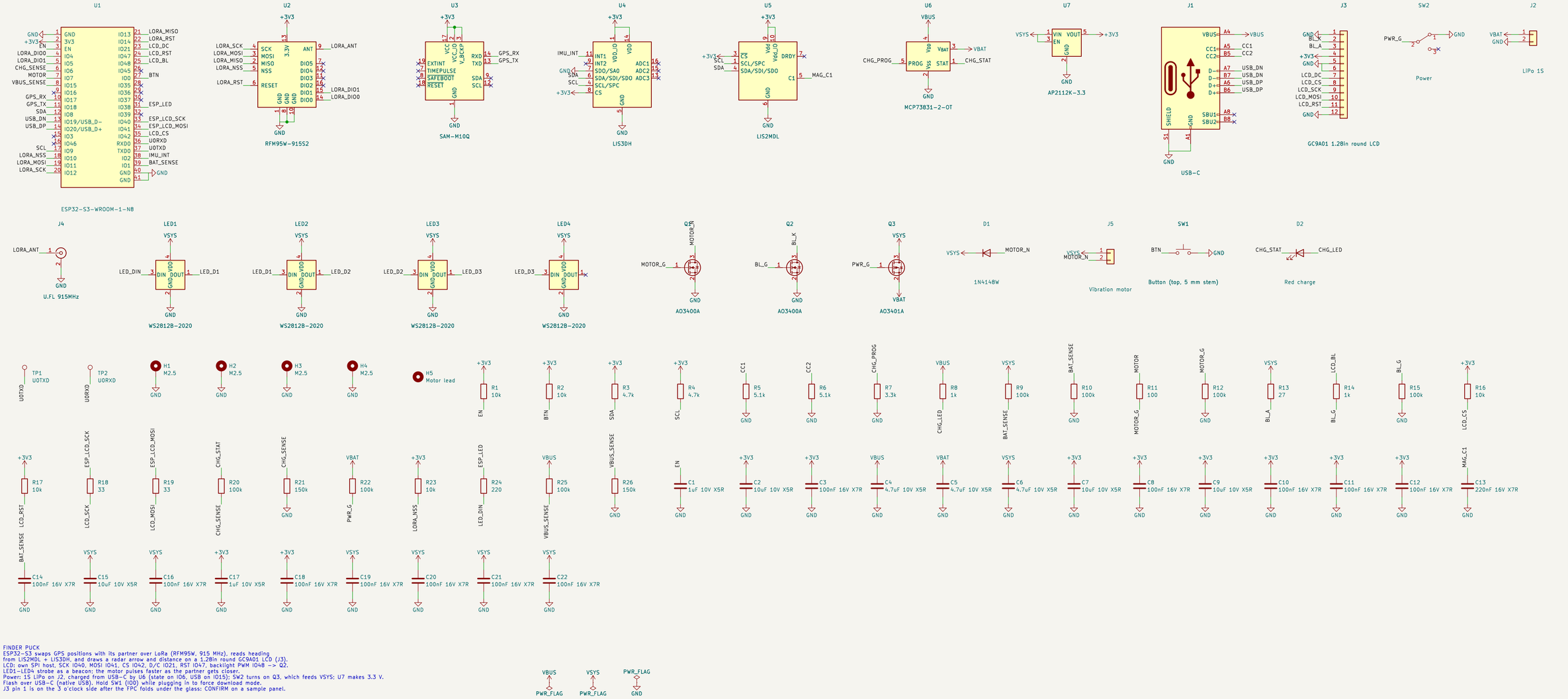


FINDER PUCK
ESP32-S3 swaps GPS positions with its partner over LoRa (RFM95W, 915 MHz), reads heading
from LIS2MDL + LIS3DH, and draws a radar arrow and distance on a 1.28in round GC9A01 LCD (J3).
LCD: own SPI host, SCK IO40, MOSI IO41, CS IO42, D/C IO21, RST IO47, backlight PWM IO48 -> Q2.
LED1-LED4 strobe as a beacon: the motor pulses faster as the partner gets closer.
Power: 1S LiPo on J2, charged from USB-C by U6 (state on IO6; USB on IO15); SW2 turns on Q3, which feeds VSYS; U7 makes 3.3 V.
Flash over USB-C (native USB): Hold SW1 (IO0) while plugging in to force download mode.
J3 pin 1 is on the 3 o'clock side after the FPC folds under the glass: CONFIRM on a sample panel.

Sheet: /	
File: finder_puck.kicad_sch	
Title: Finder Puck	
Size: A2	Date: 2026-10-19
KiCad E.D.A. kicad-cli 7.0.7	Id: 1/1

ESP32-S3-WROOM-1, RFM95W 915 MHz, SAM-M10Q, LIS3DH + LIS2MDL, GC9A01 round LCD, 1S LiPo
Two pucks point at each other: GPS + compass + LoRa, round radar screen

Rev: 0.1